GEp-4 Reboot Feasibility Study

Data anchors, present assessment, and critical-path questions

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Working configuration: 10.7 GeV beam, 75 μA, 30 cm LH_2, positive-polarity HMS proton arm, and NPS electron arm near 3 m.

What we set out to establish

Physics reach

Does the HMS–NPS configuration provide enough accepted elastic events at $Q^2 \simeq 11\ \mathrm{GeV}^2$ for a competitive form-factor measurement?

NPS response and backgrounds

Can a high NPS threshold retain elastic efficiency while controlling electron, photon, and charged-hadron backgrounds?

Trigger and DAQ

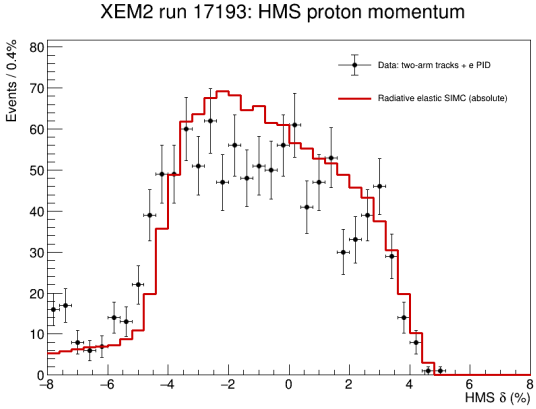
What are the real HMS and NPS singles rates, the accidental-coincidence rate, and the required Hall C DAQ capability?

Experimental design

Are the NPS distance, FPP configuration, HMS trigger logic, and coincidence timing mutually compatible?

Approach: anchor SIMC and GEANT4 to real Hall C data before extrapolating.

Elastic yield: a close HMS benchmark supports the proposal



XEM2 run 17193: HMS $+6.476$ GeV/c at 19.52°; proposed HMS $+6.735$ GeV/c at 18.80°.

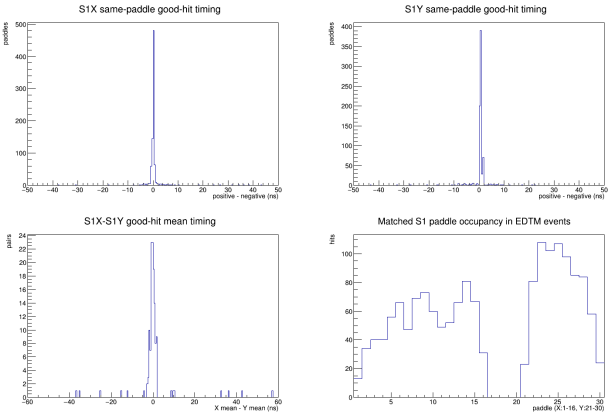
- Broad HMS acceptance: data/SIMC $= 0.936$.
- Tight elastic selection: $800/1034.34 = 0.7734$.
- At fixed 30 cm target length, NPS at 3 m gains a factor 4.11 over SHMS-like electron acceptance.
- Transfer from the XEM2 setting to $Q^2 \simeq 11\ \mathrm{GeV}^2$: SIMC ratio $= 0.8865$.
- Direct $Q^2 \simeq 11\ \mathrm{GeV}^2$ target scan: 20/10 cm $= 1.85$ and 30/10 cm $= 2.57$; the final 10 cm supplies 28% of the 30 cm yield.

45 PAC-day projection

Before NPS response	**10.69 M**
70% threshold + fiducial	**7.70 M**
80% threshold + fiducial	**7.51 M**

Response values remain provisional (3×3 energy, radiative photons, passive material). The sweeper will be absent in GEp; retaining it here is a small conservative effect.

HMS trigger: the practical S1 rate is 40–60 kHz



Run 17193 unbiased EDTM windows. A valid plane hit requires both PMTs on one physical paddle; valid X and Y paddle numbers are independent.

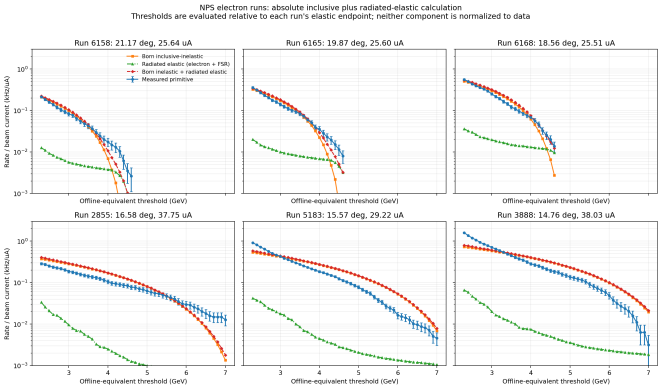
- The branch previously labeled "S1X AND S1Y" is actually **S1X OR S1Y**; its GEp-scaled rate is about 810 kHz.
- EDTM plane rates reproduce direct scalers to 3–4%.
- Requiring a double-ended paddle hit in both planes gives 12.73–14.84 kHz at the run conditions.
- Naive GEp scaling gives 51–60 kHz; scaler inclusion–exclusion gives 39.9 kHz.
- HMS 3-of-4 scales to about **2.9 kHz** if it is compatible with the FPP configuration.

Accidental scale

For 50 kHz HMS and 1 MHz NPS:

$$R_{\mathrm{acc}} = R_H R_N \Delta t = \begin{cases} 1\ \mathrm{kHz}, & 20\ \mathrm{ns} \\ 2\ \mathrm{kHz}, & 40\ \mathrm{ns}. \end{cases}$$

NPS singles: the inclusive-electron component is data anchored



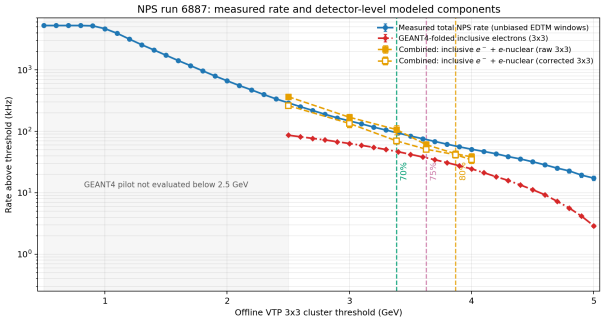
- Three-pass NPS electron data: generator/data $= 0.95$–1.35.
- Five-pass NPS electron data: trend reproduced; most discrepancies are conservative overpredictions.
- XEM2 SHMS electron-enriched trigger: 447.7 Hz predicted versus 592 Hz measured, or 74.8%.

GEp full-face prediction

Threshold	Inclusive e^-	+ radiated elastic
70%	14.34 kHz	**14.42 kHz**
75%	10.26 kHz	**10.30 kHz**
80%	7.05 kHz	**7.08 kHz**

These are genuine-electron rates, not the total NPS trigger rate.

Run 6887 establishes a large non-electron excess

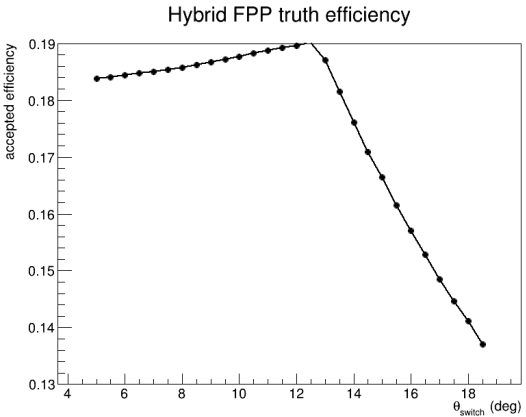


Run 6887: DVCS production, NPS on the photon arm; 8.456 GeV beam, 29.4 μA, 10 cm LH_2; NPS at 16.57° and 3.57 m. VTP single-cluster trigger/readout settings: 0.50/0.25 GeV; 8,972 unbiased 1-μs EDTM windows after the 25–35 μA current selection.

What the data establish

At 70–80%, inclusive electrons alone explain only 25–50% of the rate. Adding importance-sampled electronuclear production gives 107, 61, and 43 kHz with raw 3x3 energy, or 70, 51, and 42 kHz with the inherited correction. The six direct points use the 0.25-GeV production cut. **The background is now at the required scale**; the high-threshold deficit and 3x3 energy convention remain open.

A single analyzer preserves the small-angle FPP acceptance



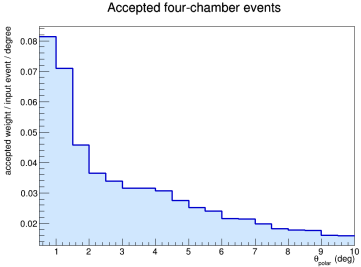
- Legacy FPP frame: one 22-inch CH_2 analyzer.
- $\theta_{\mathrm{polar}} < 12^\circ$: reconstruct with FPP1–FPP4; four chambers constrain the left/right ambiguity.
- $\theta_{\mathrm{polar}} > 12^\circ$: reconstruct with FPP1 and FPP2; the larger deflection resolves the ambiguity and avoids downstream cone-test loss.
- Truth efficiency peaks near the 12° transition.

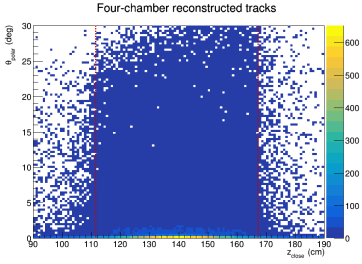
Four-chamber closure

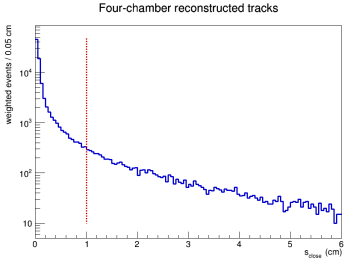
All 59,113 truth-selected events from 0° to 10° reconstruct in the ideal fit, with no large angular outliers.

Ideal closure: truth-associated hits and exact drifts; efficiency, resolution, noise, and pattern recognition remain.

Hall C chamber geometry and wire-level tracking



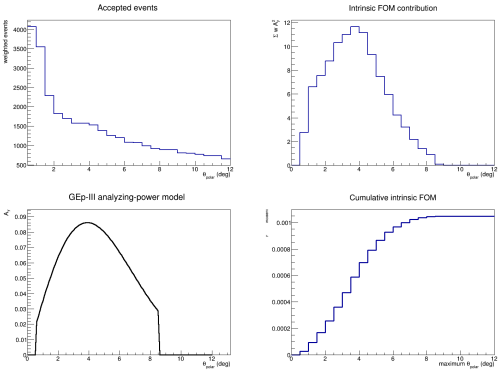




- Ported the four measured chamber positions and twelve U/X/V planes from the GEp-III parameters: 104/83/104 wires per chamber with 2 cm pitch.
- Digitize plane and wire number plus unsigned drift distance; enumerate the left/right signs and retain the minimum reduced-χ^2 $x(z), y(z)$ fit.
- Calculate the FPP observables relative to the HMS track and apply the cone test. The efficiency density at left includes the analyzer-volume, cone-test, reconstruction, and 1 cm s_{close} requirements.

Ideal closure: truth-associated hits and exact drifts; the fit receives neither the truth coordinate nor its sign. Detector response and multi-track recognition remain.

FPP acceptance and intrinsic figure of merit



6.7 GeV/c, legacy 22-inch CH_2, GEp-III A_y model, $0.07 < p_T < 1.0$ GeV/c.

The A_y parameterization comes from GEp-III/GEp-2γ and was later used in SBS planning; its use at 6.7 GeV/c is an extrapolation, not an SBS data validation. The quoted $A_{y,\mathrm{eff}} = 0.06311$ is the A_y^2-weighted RMS over accepted useful-p_T events, not the peak or an ordinary mean.

Intrinsic polarimeter result

Useful-p_T efficiency	26.264%
Effective A_y	0.06311
FOM / incident proton	0.001046
FOM / proton, $P_e = 0.85$	0.0007558

Angular content

About 76% of the FOM lies below 5°. A 1° lower cut retains 97.3% of the integrated FOM.

Tracking status

Implemented: actual chamber and wire geometry, unsigned drift distances, left/right enumeration, simultaneous $x(z)$, $y(z)$ fits, minimum-χ^2 selection, reconstructed observables, and the cone test. **Still idealized:** truth-associated hits and exact drifts; no chamber inefficiency, resolution, noise, or multi-track recognition.

FPP figure of merit and projected component precision

Define the beam-polarization-weighted FOM per incident elastic proton as

$$\mathrm{FOM} = \epsilon_{\mathrm{FPP}}\,(P_e A_y)^2, \qquad \sigma_{\mathrm{component}} = \sqrt{\frac{2}{N\,\mathrm{FOM}}}.$$

SBS planning benchmark

$$\begin{aligned}\mathrm{FOM}_{\mathrm{SBS}} &= (0.20)(0.85 \times 0.06311)^2 \\ &= 5.75 \times 10^{-4} \text{ per incident proton},\end{aligned}$$

$$\begin{aligned}\sigma_{\mathrm{component}} &= \sqrt{\frac{2}{(9.3 \times 10^6)(5.75 \times 10^{-4})}} \\ &\simeq 0.0193.\end{aligned}$$

Current Hall C projection

$$\mathrm{FOM}_{\mathrm{Hall\ C}} = 7.558 \times 10^{-4} \text{ per incident proton},$$

$$N = 7.5\text{–}7.7 \times 10^6,$$

$$\begin{aligned}\sigma_{\mathrm{component}} &= \sqrt{\frac{2}{N\,\mathrm{FOM}_{\mathrm{Hall\ C}}}} \\ &= 0.0188\text{–}0.0185.\end{aligned}$$

Interpretation

FOM is a dimensionless weight per incident proton; $N\,\mathrm{FOM}$ is the total statistical weight for one polarization component. It is **not** the uncertainty in $\mu_p G_E/G_M$. For SBS, the 20% efficiency is an assumed effective reconstructed efficiency.